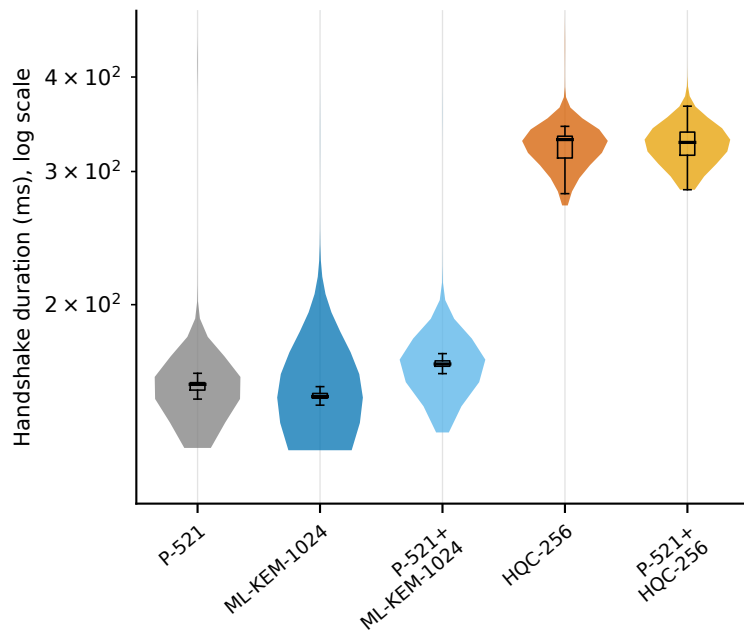
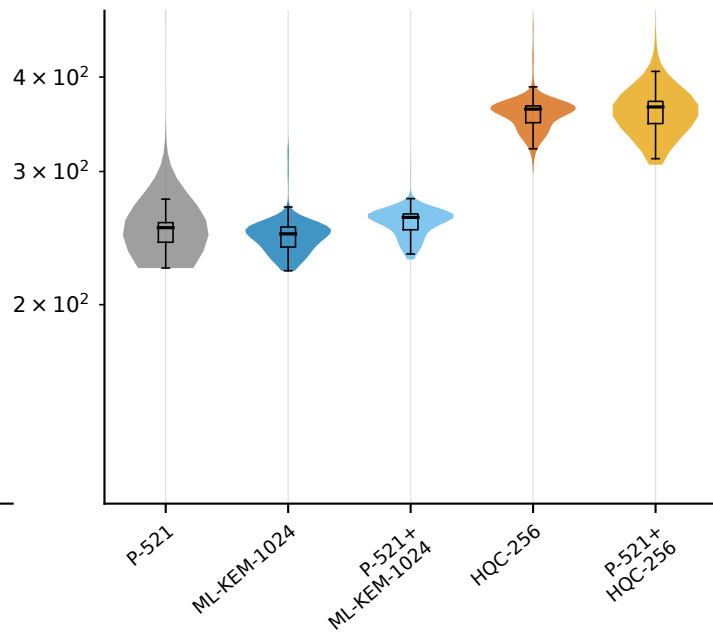


(a)

secp521r1 (classical)

**(b)**

mldsa87 (post-quantum)

**(c)**

Classical KEX
 ML-KEM (pure)
 ML-KEM (hybrid)
 HQC (pure)
 HQC (hybrid)

KEM	secp521r1 Mean (ms)	secp521r1 CV (%)	secp521r1 Max (ms)	mldsa87 Mean (ms)	mldsa87 CV (%)	mldsa87 Max (ms)
P-521	157.4	31.8	1.16k	257.0	33.9	1.39k
ML-KEM-1024	159.4	59.3	1.23k	247.6	8.5	515.0
P-521+ML-KEM-1024	169.6	30.6	1.25k	258.4	7.6	536.2
HQC-256	327.6	16.7	1.43k	360.6	8.0	653.8
P-521+HQC-256	329.7	17.0	1.44k	366.6	20.7	1.52k